Data-Driven Strategies for Targeted Vaccine Outreach

VACCINES PREDICTION ANALYSIS

BY: ANGELA WANGUI WACHIRA

Project Overview

- Vaccines are essential for controlling infectious diseases and achieving herd immunity. This project needs to use machine learning to predict uptake of H1N1 and seasonal vaccines by analyzing demographic, behavioral, and attitudinal data. The findings identify key factors influencing vaccination decisions and help improve future public health strategies and responses to future infectious disease outbreaks.

BUSINESS PROBLEM

Public health organizations struggle to achieve high vaccination rates due to differences in beliefs, behaviors, and healthcare access. This project predicts the likelihood of individuals receiving H1N1 and seasonal vaccines by estimating two probabilities.

The goal is to identify those less likely to be vaccinated and understand influencing factors, enabling targeted interventions to improve vaccine uptake and reduce disease spread.

PROJECT STAKEHOLDERS

- Public Health Agencies: Design vaccination campaigns and policies
- Healthcare Providers: Identify and encourage at-risk patients
- Researchers & Data Scientists: Analyze trends and improve predictive models

DATA UNDERSTANDING

The dataset contains 26,707 survey responses collected during the 2009 H1N1 pandemic, with 36 features categorized into behavioral, demographic, and attitudinal groups. It includes information on protective actions, personal characteristics, and beliefs about the flu and vaccines.

The target variables are binary indicators showing whether individuals received the H1N1 and seasonal vaccines.

My Target Variables

The two binary labels we want to predict:

h1n1_vaccine

seasonal_vaccine

MODELLING APPROACH

Baseline: Logistic Regression

Establishes fast, interpretable benchmark for vaccine uptake

SMOTE Logistic Regression

Prioritize Class 1 Recall to identify vaccinated individuals

Decision Tree

Explore non-linear relationships and interactions

Subcutaneous VRandom Forest (Tuned)

Optimizes performance via ensemble learning

Model Performance Comparison

H1N1 VACCINE

Winning Model: SMOTE Logistic Regression

- Class 1 Recall: 71% (highest among all models)
- ROC-AUC: 0.78
- Accuracy: 78%

SEASONAL VACCINE

Winning Model: Random Forest

- Class 1 Recall: 74%
- ROC-AUC: 0.83
- Accuracy: 78%

The Random Forest demonstrated maximum algorithmic stability for the balanced seasonal dataset, while SMOTE Logistic Regression excelled at probability threshold management for the imbalanced H1N1 data.

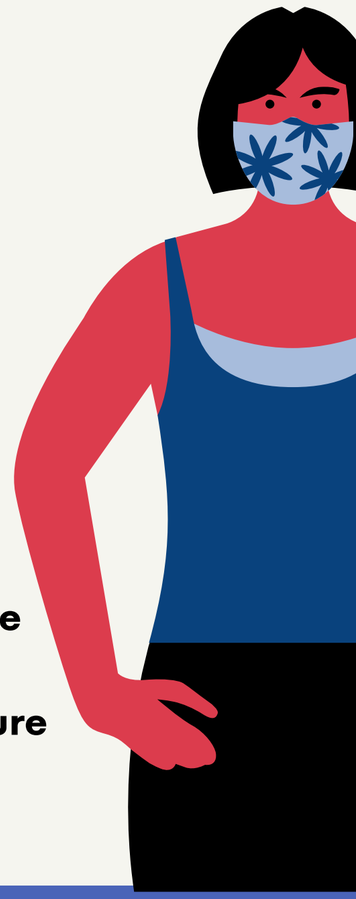
Key Factors Affecting Vaccine Uptake

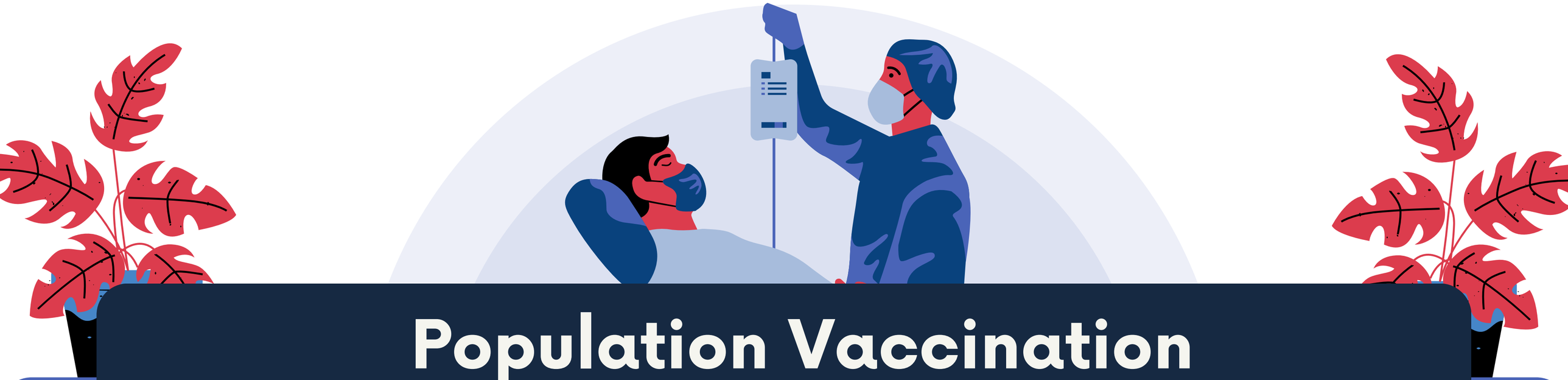
- **Doctor's Recommendation- The single most influential factor for both H1N1 and seasonal vaccines. Direct physician outreach carries the highest mathematical weight across all models.**

- **Perceived Effectiveness- Individuals' belief in vaccine effectiveness significantly increases probability of H1N1 uptake. Effectiveness perception is second only to doctor recommendation.**

- **Personal Risk Perception- Perceived personal risk of contracting the virus drives both vaccines. For seasonal flu, age 65+ becomes a massive demographic indicator.**

- **Healthcare Worker Status- Being a healthcare worker significantly increases H1N1 vaccine probability, likely due to occupational exposure and education.**





Population Vaccination Probabilities

17.59%

Both Vaccines
Received H1N1 and seasonal flu vaccines

50.22%

Either Vaccine
Received at least one of the two vaccines

49.78%

Neither Vaccine
Did not receive either H1N1 or seasonal vaccine

Effectiveness Comparison

Seasonal Vaccine Uptake

Higher public reach and adoption rate

H1N1 Vaccine Uptake

Lower reception compared to seasonal flu

The seasonal vaccine has been significantly more effective in terms of public reach and adoption. The Logistic Regression model with SMOTE proved most effective for identifying the harder-to-reach H1N1 vaccinated group.

Final Recommendations

Deploy Random Forest for seasonal optimization.

Allocate a majority of educational campaign budgets specifically toward H1N1 normalization.

Begin capturing physical geographic features to improve future models.

Future Iterations (Next Steps)

- **Advanced Modeling: Transition to Gradient Boosted Trees (XGBoost) to capture non-linear interactions.**

- **Geospatial Data: Adding zip-code level data to account for "Pharmacy Deserts" or proximity to healthcare.**

- **Temporal Analysis: Applying this model to COVID-19 or current flu data to see if behavior patterns have shifted since 2**

Questions or comments?

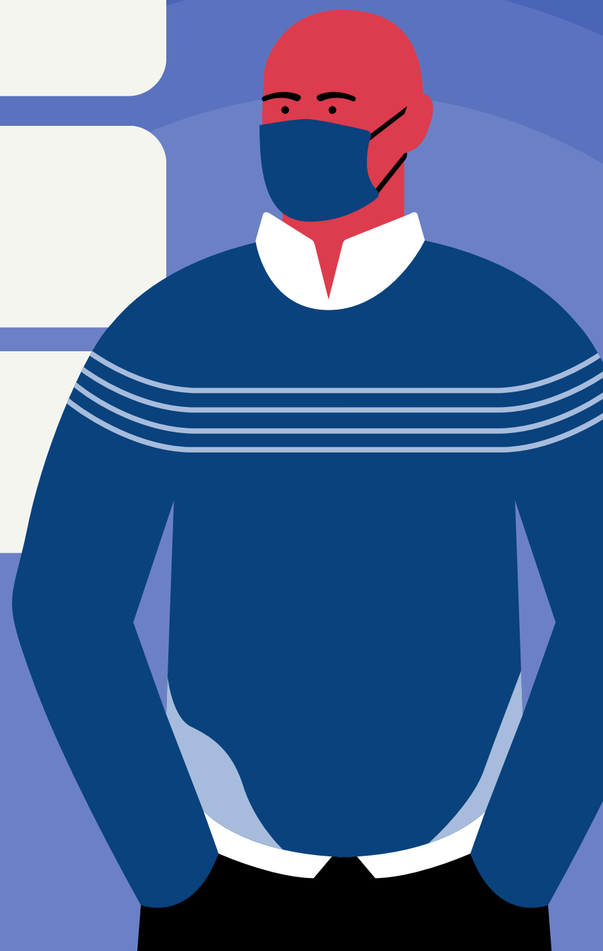
Get in touch!

Phone Number

0795799763

Email Address

angela.wachira@student.moringaschool.com



THANK YOU